

BCD Counter (0–9) with Adjustable Clock and Seven-Segment Display

1. Objective

The objective of this project is to design and implement a digital counter that counts from **0 to 9 (BCD)** and displays the result on a **seven-segment display**. A **NE555** is used to generate a variable clock signal, a **74LS90N** performs the counting operation, and a **CD4511** drives the display.

A **potentiometer** is used to control the speed of the counter by adjusting the clock frequency.

2. Components Used

Component	Function
NE555 Timer	Generates adjustable clock pulses
74LS90	BCD decade counter (0–9)
CD4511	BCD to seven-segment decoder
7-Segment Display (Common Cathode)	Displays digits
Potentiometer	Controls clock speed
Resistors	Current limiting
Capacitor	Sets timing
Breadboard & wires	Circuit construction

3. System Overview

The system consists of three main blocks:

NE555 Clock → 74LS90 Counter → CD4511 Decoder → 7-Segment Display

- The NE555 produces a periodic clock signal
 - The 74LS90 counts each pulse in BCD
 - The CD4511 converts BCD into segment signals
 - The display shows digits from 0 to 9 repeatedly
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4. 555 Timer Design (Astable Mode)

The NE555 is configured as an **astable multivibrator**, meaning it continuously generates pulses.

Potentiometer Configuration

- Pins **1 and 2 of the potentiometer** are tied together
- This node connects to **pin 7 (discharge) of the NE555**
- The **wiper (pin 3)** connects to **pin 2 (trigger)** of the NE555

This configuration allows adjustment of the **charging/discharging path**, thus controlling frequency.

Frequency Equation

$$f = 1.44 / (R1 + 2R2)C$$

The potentiometer effectively varies $R2$, changing the oscillation speed.

5. Counter Design (74LS90)

The 74LS90 is configured as a **decade (mod-10) counter**.

Internal Operation

- Divides clock pulses into BCD output
- Counts from **0000 to 1001 (0 to 9)**
- Automatically resets after 9

Output Bit Weights

Output	Weight
QA	1
QB	2
QC	4
QD	8

6. BCD Truth Table

Decimal	QD	QC	QB	QA	Binary
0	0	0	0	0	0000
1	0	0	0	1	0001
2	0	0	1	0	0010
3	0	0	1	1	0011
4	0	1	0	0	0100
5	0	1	0	1	0101
6	0	1	1	0	0110
7	0	1	1	1	0111
8	1	0	0	0	1000
9	1	0	0	1	1001

7. Decoder Logic (CD4511)

The CD4511 converts BCD input into seven-segment outputs.

Control Pins

Pin	Function	Setting
LT	Lamp Test	HIGH
BI	Blanking Input	HIGH
LE	Latch Enable	LOW

These settings allow real-time display updates.

8. Seven-Segment Display Truth Table

Digit	a	b	c	d	e	f	g
0	1	1	1	1	1	1	0
1	0	1	1	0	0	0	0
2	1	1	0	1	1	0	1
3	1	1	1	1	0	0	1
4	0	1	1	0	0	1	1
5	1	0	1	1	0	1	1
6	1	0	1	1	1	1	1
7	1	1	1	0	0	0	0
8	1	1	1	1	1	1	1
9	1	1	1	1	0	1	1

9. Circuit Operation

1. The NE555 generates clock pulses at a frequency controlled by the potentiometer
2. Each pulse increments the 74LS90 counter
3. The counter outputs a 4-bit BCD number
4. The CD4511 decodes this into segment signals
5. The seven-segment display shows the corresponding digit

The process repeats continuously from **0 to 9**.

10. Design Considerations

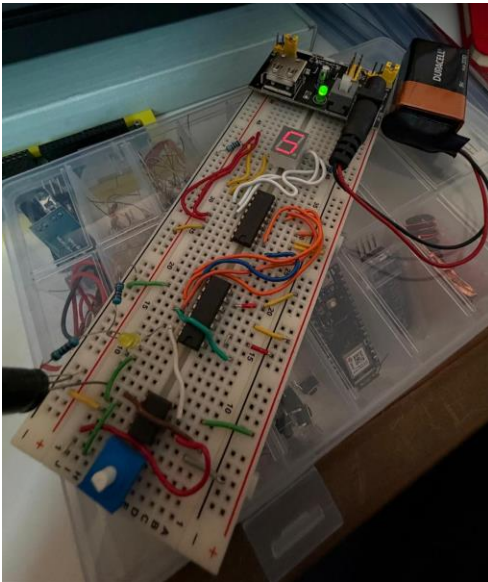
- The CD4511 requires a **common cathode display**
 - Each segment must include a **current-limiting resistor ($\sim 330\Omega$)**
 - All ICs must share a **common ground**
 - Stable power supply (5V) is required for proper operation
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11. Conclusion

This project demonstrates the integration of analog and digital components to create a functional counting system. The NE555 provides a controllable clock source, the 74LS90 performs sequential counting, and the CD4511 enables human-readable output through a seven-segment display. The addition of a potentiometer allows dynamic control of counting speed, enhancing usability and demonstrating real-time system control.

12. Notes for Submission

Breadboard photo



Video:

Schematic diagram

